Eleni Tzirita Zacharatou
Curriculum Vitae

Associate Professor, Hasso Plattner Institute
Campus II, F-E.03, August-Bebel-Str. 88, D-14482 Potsdam, Germany
eleni.tziritazacharatou@hpi.de | ORCID 0000-0001-8873-5455 | heltzi.github.io

---

Education

Ph.D. in Computer Science — 2013 – 2019
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Thesis: Efficient Query Processing for Spatial and Temporal Data Exploration
Advisor: Prof. Anastasia Ailamaki

Diploma (M. Eng.) in Electrical & Computer Engineering — 2007 – 2013
National Technical University of Athens (NTUA), Greece
Thesis: Automatic Music Transcription
Advisor: Prof. Petros Maragos

Academic Experience

Hasso Plattner Institute (HPI), University of Potsdam — Potsdam, Germany
Associate Professor (W2, tenure track) — March 2025 –
- Lead of Research Group *Spatial Analytics and Large-Scale Data Processing*

Berlin Institute for the Foundations of Learning and Data (BIFOLD) — Berlin, Germany
Junior Fellow — July 2021 – now

IT University of Copenhagen (ITU) — Copenhagen, Denmark
Assistant Professor — May 2022 – February 2025
- Member of the Data-Intensive Systems and Applications (DASYA) group
- Research Focus: Spatial Big Data Analytics

Technical University of Berlin (TUB) — Berlin, Germany
Postdoctoral Researcher, Computer Science Department — Oct 2019 – Apr 2022
- Member of the Database Systems and Information Management (DIMA) group
- Supervisor: Prof. Volker Markl

New York University (NYU) — New York, USA
Visiting Researcher, Tandon School of Engineering — Jul 2016 – Oct 2016
- Member of the Visualization, Imaging and Data Analysis (VIDA) group
- Supervisors: Prof. Juliana Freire and Dr. Harish Doraiswamy

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Doctoral Assistant, Computer Science Department — Sep 2013 – Aug 2019
- Member of the Data-Intensive Applications and Systems (DIAS) group
- Research Focus: Spatial and temporal data management for exploratory applications
- Advisor: Prof. Anastasia Ailamaki

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Research Intern, Computer Science Department Jan 2013 – Aug 2013
- Member of the Data-Intensive Applications and Systems (DIAS) group
- Supervisors: Prof. Anastasia Ailamaki and Prof. Thomas Heinis

Honors and Awards

- Distinguished Reviewer Award, VLDB Conference 2025
- Distinguished Reviewer Award, VLDB Conference 2022
- Junior Fellowship, Berlin Institute for the Foundations of Learning and Data (BIFOLD) 2021
- Best Demonstration Award, ACM SIGMOD Conference 2018
- Travel Award, ExploreDB Workshop (in conjunction with ACM SIGMOD), USD 1K 2016
- Distinction in the Panhellenic Math Competition "Euclid", Hellenic Mathematical Society 2006

Publications Summary

Full papers at top-tier conferences (CORE A*/A ranking) and journals: 14
Other peer-reviewed conference papers: 6
Peer-reviewed workshop and demonstration papers: 13
Number of citations: 536 and h-index: 12 (as per Google Scholar in September 2026)
Full list of publications: see https://heltzi.github.io/publications.html

Invited Talks and Conference Presentations

- 09/2026: Adaptive Data Systems for a Heterogeneous World, Chalmers University of Technology, Gothenburg, Sweden
- 09/2026: Adaptive Data Systems for a Heterogeneous World, Inaugural Lecture (Antrittsvorlesung), HPI, Potsdam, Germany
- 07/2026: Nova: Scalable Streaming Join Placement and Parallelization in Resource-Constrained Geo-Distributed Environments, Hellenic Data Management Symposium (HDMS), Athens, Greece
- 02/2026: Where to Aggregate? Scalable Operator Placement for Stream Processing from Edge to Cloud, Humboldt University, Berlin, Germany
- 12/2025: Exploiting Heterogeneity in Modern Data Systems, EPFL, Lausanne, Switzerland
- 09/2025: Spatial Data Systems, Readings in Data Systems Seminar, ITU, Copenhagen, Denmark
- 06/2025: City data speaks – but how do we listen?, Soapbox Science Berlin, Holzmarkt, Germany
- 06/2025: Efficient Data Management for Spatial and IoT-Driven Applications, Snowflake, Berlin, Germany
- 05/2025: Efficient Data Management for Spatial and IoT-Driven Applications, Data & AI Cluster Meeting, HPI, Germany
- 03/2025: Efficient Aggregation in a Heterogeneous World, Next-Generation Data Management Systems event by Huawei@EDBT, Barcelona, Spain
- 03/2025: "Meet Your Mentor" Short Introduction, AI Grid, Online
- 10/2024: Unifying Spatial Data Analytics, Data Management Aspects of Analytics session at D3A Conference, Nyborg, Denmark
- 07/2024: Spatial Data Management Systems, VLDB Summer School, Cluj-Napoca, Romania
- 07/2024: Surviving in Academia: Personal Story and Advice, BIFOLD Career Seminar, Berlin, Germany
- 07/2024: Transforming Heterogeneous Spatial Data into Insights, BIFOLD Lunch Talk, Berlin, Germany
- 06/2024: Analysis of Geospatial Data Loading, DBTest Workshop@SIGMOD, Santiago, Chile
- 11/2023: Efficient Data Management for Modern Spatial Applications, HPI-TU Darmstadt Joint Database Systems Seminar, Online
- 09/2023: Efficient Data Management for Modern Spatial Applications, NERDS (ITU) Seminar, Denmark
- 09/2022: Efficient Data Management for Modern Spatial Applications and the Internet of Moving Things, DIREC Seminar, Aarhus, Denmark

- 09/2021: Agora-EO: A Unified Ecosystem for Earth Observation, Monthly Research Meeting, Department of Geo-information Processing, University of Twente, Online
- 01/2021: The Case for Distance-Bounded Spatial Approximations, CIDR Conference, Online
- 11/2019: Efficient Bundled Spatial Range Queries, SIGSPATIAL Conference, Chicago, USA
- 08/2019: Efficient Query Processing for Spatial and Temporal Data Exploration, Public Defense, EPFL, Lausanne, Switzerland
- 07/2019: GPU Rasterization for Real-Time Spatial Aggregation over Arbitrary Polygons, Hellenic Data Management Symposium (HDMS), Athens, Greece
- 02/2019: Interactive and Exploratory Spatio-Temporal Data Analytics, DIMA Research Seminar, TU Berlin, Germany
- 08/2018: GPU Rasterization for Real-Time Spatial Aggregation over Arbitrary Polygons, VLDB Conference, Rio de Janeiro, Brazil
- 06/2018: Interactive Exploration of Urban Data with GPUs, Short Presentation, IC Research Day, EPFL, Lausanne, Switzerland
- 04/2018: Improving Spatial Data Processing by Clipping Minimum Bounding Boxes, ICDE Conference, Paris, France
- 06/2017: Real-Time Spatial Aggregation using GPU Rasterization, EcoCloud Annual Event, Lausanne, Switzerland
- 09/2016: Indexing the brain, Data Science Lunch Talk, NYU Center for Data Science, New York, USA
- 06/2015: RUBIK: Efficient Threshold Queries on Massive Time Series, SSDBM Conference, San Diego, USA

Panels

- Diversity, Equity, and Inclusion Panel, Panel Moderator, EDBT, March 2025
- Ph.D. Mentoring Panel, VLDB, August 2024
- A Discussion on Success in a Ph.D. Journey, VLDB Ph.D. Workshop, August 2024
- Spatial and Temporal Data Management Panel, ICDE, May 2022
- Spatio-Temporal Data System Panel, VLDB, August 2021

Advising (Current)

- *Rosmarie Debski, Jan Kubeler, Joshua Storost, Luca Vannuccini*, Master's Project "Querying Earth from Orbit", HPI — Fall 2026 – now
- *Maximilian Schulze*, PhD student, HPI — Fall 2025 – now
- *Huan Zhang*, PhD student, HPI — Spring 2026 – now
- *Anton Hackl*, Research Assistant (M.Sc. student), HPI — Fall 2025 – now
- *Felix Treykorn*, Research Assistant (M.Sc. student), HPI — Spring 2026 – now
- Mentor in the AI Grid network — 2023 – now

Advising (Alumni)

- *Gereon Dusella*, PhD, TUB, Primary advisor: Prof. Volker Markl — Fall 2023 – Spring 2026
- *Xenofon Chatziliadis*, PhD, TUB, Primary advisor: Prof. Volker Markl — Fall 2019 – Fall 2025
- *Martin Kiefer*, PhD, TUB, Primary advisor: Prof. Volker Markl — Fall 2021 – Fall 2022
- *Felix Treykorn, Anton Persitzky, Maximilian Speer*, Master's Project, HPI, — Fall 2025 – Spring 2026
- *Aske Wachs*, MSc Research Project & Thesis, ITU — Fall 2023 – Spring 2024
- *Johan Kjersgård Flensmark, Benjamin Hammer Thygesen, Alexander Theodor Bilde Pedersen*, MSc Research Project & Thesis, ITU — Fall 2023 – Spring 2024
- *Barbara Puljak*, MSc Research Project & Thesis, ITU — Fall 2023 – Spring 2024
- *Dagmara Kamila Przygocka, Kamila Garczynska*, MSc Research Project, ITU — Fall 2023
- *Thanasis Georgiadis*, Research Mentor, UoI, Primary advisor: Prof. Nikos Mamoulis — Spring 2023
- *Hanno Skowronek*, MSc Thesis, TUB, Primary advisor: Prof. Begüm Demir — Spring 2023

- *Carl Bruun & Rasmus Garbarsch*, BSc Thesis, ITU — Spring 2023
- *Gergo Gyori*, BSc Thesis, ITU — Spring 2023
- *Gereon Dusella*, MSc Thesis, TUB, Co-supervisor: H. Gavriilidis — Spring 2023
- *Peter Løffmann, Thomas Haugaard Mogensen, Zachery Kuntz*, MSc Research Project, ITU — Fall 2022
- *Felix Henze*, Research Mentor, TUB, Co-supervisor: H. Gavriilidis — Spring 2022
- *Ahmet Kerem Aksoy*, Research Mentor, TUB — Spring 2022
- *Laert Nuhu*, MSc Thesis, TUB, Co-supervisor: H. Gavriilidis — Spring 2022
- *Christian Winter*, Research Mentor, TUM, Primary advisor: Prof. Alfons Kemper — Fall 2021
- *Pablo Alberto Osinaga Arze*, MSc Thesis, TUB — Fall 2021
- *Iva Mihajlovska*, MSc Thesis, TUB — Spring 2021
- *Marcellus Prama Saputra*, BSc Thesis, TUB — Spring 2021
- *Robin Papke*, MSc Thesis, TUB — Fall 2020
- *Elena Beatriz Ouro Paz*, MSc Thesis, TUB — Spring 2020
- *Carlos E. Muniz Cuza*, MSc Thesis, TUB & Aalborg University, Co-supervisors: Professors Nguyen Thi Thao Ho, Torben Bach Pedersen, Bin Yang — Spring 2020
- *Vlad Mihaescu*, MSc Research Project, EPFL — Fall 2018
- *Maria Ilina*, Summer Intern, EPFL — Summer 2018
- *Parand Alizadeh*, Summer Intern, EPFL — Summer 2018
- *Chenxi Sun*, Summer Intern, EPFL — Summer 2015

PhD Thesis Committees

- Jingyu Liu, Licentiate Seminar, Discussion Leader, Chalmers (Advisor: Vincenzo Gulisano) — 2026
- Marcus Pappik, PhD Defense, Examiner, HPI (Advisor: Tobias Friedrich) — 2026
- Mariana Garcez Duarte, PhD Defense, Examiner, ULB (Advisors: Mahmoud Sakr and Esteban Zimanyi) — 2026
- Wang Yue, PhD Defense, Examiner, HPI (Advisor: Tilmann Rabl) — 2025
- Sebastian Schmidl, PhD Defense, Examiner, HPI (Advisor: Felix Naumann) — 2025
- Xenofon Chatziliadis, PhD Defense, Reviewer, TU Berlin (Advisor: Volker Markl) — 2025
- Thanasis Georgiadis, PhD Defense, Reviewer, University of Ioannina (Advisor: Nikos Mamoulis) — 2025
- Keven Richly, PhD Defense, Reviewer, HPI (Advisor: Hasso Plattner) — 2024
- Martin Kiefer, PhD Defense, Reviewer, TU Berlin (Advisor: Volker Markl) — 2023
- Morten Tychsen Clausen, PhD Midway, ITU (Advisor: Philippe Bonnet) — 2023

Teaching

Term	Institute	Course Name	Role
Winter 2026/2027, Winter 2025/2026	HPI	Spatial Databases Lecture (6 ECTS, MSc)	Course Manager & Responsible Lecturer
Winter 2026/2027	HPI	Querying Earth from Orbit: Distributed Spatial Query Processing Across Low-Earth Orbit (LEO) Satellite Constellations (Master's Project, 12 ECTS)	Project Mentor
Spring 2026	HPI	Management of Complex Data Types (6 ECTS, MSc)	Course Manager & Responsible Lecturer

Winter 2025/2026	HPI	Lecture Series on Database Research (3 ECTS)	Lecturer
Winter 2025/2026	HPI	Enabling Cross-Platform Spatial Analytics with Apache Wayang (Master's Project, 12 ECTS)	Project Mentor
Spring 2025	HPI	Spatial Data Management Project Seminar (6 ECTS, MSc)	Course Manager & Responsible Lecturer
Fall 2023 – Fall 2024 & Fall 2022	ITU	Introduction to Database Systems	Course Manager & Responsible Lecturer
Fall 2021, Spring 2021	TUB	DBPRA: Datenbankpraktikum (Practical course on databases)	Responsible Lecturer
Fall 2020	TUB	AIM-3: Scalable Data Analysis and Data Mining	Lecturer
Fall 2020, Spring 2020, Fall 2019	TUB	BDASEM: Big Data Analytics Seminar	Lecturer
Spring 2020	TUB	BDAPRO: Big Data Analytics Project	Project Mentor
Fall 2019	TUB	DBSEM: Foundations of Database Systems Seminar	Lecturer
Fall 2018/2019, Spring 2014	EPFL	Information, Computation, Communication	Teaching Assistant
Spring 2018, Spring 2017	EPFL	Database Systems	Teaching Assistant
Fall 2017/2018	EPFL	Programming I (in C)	Teaching Assistant
Spring 2016	EPFL	Probability and Statistics	Teaching Assistant
Fall 2015/2016	EPFL	Computer Architecture	Teaching Assistant
Spring 2015	EPFL	Introduction to Database Systems	Teaching Assistant
Fall 2014/2015	EPFL	Introduction to Object Oriented Programming (in Java)	Teaching Assistant

Service

Leadership & Organization:

- Treasurer, Hellenic ACM SIGMOD Chapter 2026 – 2028
- Program Co-chair, DOLAP Workshop 2026 – 2027
- Executive Guest Editor (with Alejandro Maté), Special Issue on the Best Papers of DOLAP 2026, *Information Systems* (Elsevier) 2026 – 2027
- Diversity and Inclusion Co-chair, EDBT 2025 – 2026
- Program Co-chair, Hellenic Data Management Symposium (HDMS) 2025
- Proceedings Co-chair, SIGMOD 2025

- Student Travel Grant Selection Committee Member, SIGMOD 2025
- Co-organizer, Data Management Aspects of Analytics Workshop (in conjunction with the D3A Conference) 2024
- Proceedings Chair, DEBS 2022
- Co-chair, BOSS Workshop (in conjunction with VLDB) 2020

Program Committee:
- IEEE ICDE: 2027 (Senior PC), 2024 (Demo track)
- ACM SIGMOD: 2027, 2026 (Round 4), 2025, 2024, 2023, 2022, 2021 (Industrial track)
- VLDB: 2026, 2025, 2023, 2022
- ACM SIGSPATIAL PhD and Graduate Studies Workshop & Student Research Competition: 2025
- EDBT: 2022
- VLDB PhD Workshop: 2023
- BTW: 2023 (Demo track)
- ACM SIGSPATIAL Workshop on Location-based Recommendations, Geosocial Networks and Geoadvertising (LocalRec): 2023
- PLDI Workshop on Distributions, Relational Algebra, Graphs, Semi-Rings, Tensors, and All That (DRAGSTERS): 2023
- IEEE Intl. Workshop on Testing Distributed Internet of Things Systems (TDIS): 2025, 2022
- AI-ML Systems: 2021
- CIKM: 2018 (Short research papers track)

Journal Reviewer:
- VLDB Journal, 2021 – 2022
- Knowledge and Information Systems (KAIS) Journal, 2021 – 2022
- Big Data Research Journal, 2022

External Reviewer:
CIDR 2022, SIGMOD 2021, CIDR 2021, ICDE 2017

Session Chair:
EDBT 2023, VLDB 2021, VLDB 2020

Assessment and Admission Committees:
- Hiring Committee Member at HPI: Open Topic Professorship, 2027
- Hiring Committee Member at HPI: Two W3 Professorships in Artificial Intelligence, 2026
- Hiring Committee Member at University of Hamburg: Professorship of "W3 Data Engineering", 2024
- Program Admission Committee Member at ITU: Data Science Bachelor (Spring 2024), Software Design Master (Spring 2023)
- PhD Assessment Committee Member at ITU: Spring 2024, Spring 2023, Fall 2022
- Hiring Committee Member at TU Berlin: Professorship of "W3 Big Data Engineering", 2021

Department Level:
- Co-organizer: CS Lectures Series, ITU, 2022 – 2024
- Co-organizer: CS Department Annual Retreat, ITU, 2023

Media, Blogs, and Interviews
- 2026: *Prof. Eleni Tzirita Zacharatou delivers inaugural lecture at HPI*, HPI website
- 2025: *HPI appoints BIFOLD fellow Eleni Tzirita Zacharatou as professor*, BIFOLD website
- 2025: *Newly Appointed*, UP website

- 2025: *Press release: Two new professors at the Hasso Plattner Institute*, HPI website
- 2023: *An Interview with Eleni Tzirita Zacharatou*, DIMA TUB website
- 2021: *Advancing Big Data Analytics: Post-Doctoral Research*, DataTalksClub podcast
- 2021: *BIFOLD welcomes the first six junior fellows*, BIFOLD website
- 2021: *The Case for Distance-Bounded Spatial Approximations*, The Agora Technology Blog, Medium
- 2018: *Student-teacher Duo from EPFL Wins Best Demonstration Award at SIGMOD*, Ecocloud News
- 2018: *Tandon Team Takes Home Top Honors at Association for Computing Machinery Conference*, Tandon School of Engineering News

References
Provided upon request.

Languages
Greek (Native), English (Fluent), French (Fluent), German (Intermediate)

Extracurricular
I have formally studied the piano, the violin, and music theory. I enjoy playing music alone or in groups.
- 2010: Piano Teaching Diploma (Ptychio), Grade Excellent, Athenaeum Conservatory
- 2005: Classical Harmony Diploma (Ptychio), Heraklion Conservatory

Last update: October 26, 2026.

Last update: October 26, 2026.

Eleni Tzirita Zacharatou
Publications

Associate Professor, Hasso Plattner Institute
eleni.tziritazacharatou@hpi.de | ORCID 0000-0001-8873-5455 | heltzi.github.io

Metric	Value
Peer-reviewed publications	33
Citations	536
h-index	12
i10-index	17

Table 1: Summary of bibliometrics as per Google Scholar in September 2026.

Note: The database research community is relatively small, resulting in correspondingly smaller indices for its researchers, as evidenced by conference indices.[1] The venue ranking according to the CORE Conference Portal[2] is included when available, referencing the ranking cycle valid at the time of submission. My name is highlighted in **boldface** and the names of students under my (co-)mentorship are marked with an asterisk (*).

Peer-Reviewed Journal Publications

[1] T. Georgiadis, E. **Tzirita Zacharatou**, and N. Mamoulis. "Raster Interval Object Approximations for Spatial Intersection Joins". In: *The VLDB Journal* 34(8) (2025). [Core A*].

[2] X. Chatziliadis*, E. **Tzirita Zacharatou**, A. Eracar, S. Zeuch, and V. Markl. "Efficient Placement of Decomposable Aggregation Functions for Stream Processing over Large Geo-Distributed Topologies". In: *Proceedings of the Very Large Databases Endowment (PVLDB)* 17(6) (2024), pp. 1501–1514. [Core A*].

[3] H. Gavriilidis, F. Henze*, E. **Tzirita Zacharatou**, and V. Markl. "SheetReader: Efficient Specialized Spreadsheet Parsing". In: *Information Systems* 115 (2023), p. 102183. [Core A*].

[4] M. Kiefer*, I. Poulakis, E. **Tzirita Zacharatou**, and V. Markl. "Optimistic Data Parallelism for FPGA-Accelerated Sketching". In: *Proceedings of the Very Large Databases Endowment (PVLDB)* 16(5) (2023), pp. 1113–1125. [Core A*].

[5] E. **Tzirita Zacharatou**, H. Doraiswamy, A. Ailamaki, C. T. Silva, and J. Freire. "GPU Rasterization for Real-Time Spatial Aggregation over Arbitrary Polygons". In: *Proceedings of the Very Large Databases Endowment (PVLDB)* 11(3) (2017), pp. 352–365. [Core A*].

Peer-Reviewed Conference Publications

[6] X. Chatziliadis, E. **Tzirita Zacharatou**, S. Akili, A. Eracar, and V. Markl. "Nova: Scalable Streaming Join Placement and Parallelization in Resource-Constrained Geo-Distributed Environments". In: *International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 2026, pp. 433–446. [Core2023 A].

[7] A. Hackl* and E. **Tzirita Zacharatou**. "Pierce: GPU Ray Tracing for Spatial Joins over Complex 3D Data". In: *International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. ACM, 2026. [Core A].

[1] https://scholar.google.de/citations?view_op=top_venues&hl=en&vq=eng_databasesinformationsystems
[2] www.core.edu.au/conference-portal

[8] P. Walther, F. Deuser, C. Pugliese, W. Mansour, X. Luo, J. M. Zollner, K. Anders, E. **Tzirita Zacharatou**, C. Renso, V. Hristidis, A. Eldawy, and M. Werner. "Envisioning Unified Spatial Benchmarks". In: *International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. ACM, 2026. Vision paper. [Core A].

[9] G. Dusella*, H. Gavriilidis, B. Chen, B. Demir, V. Markl, and E. **Tzirita Zacharatou**. "Automating Earth Observation Analytics Pipelines with Agent Raven". In: *Conference on Big Data from Space*. 2025, pp. 105–108.

[10] V. Pandey, A. van Renen, E. **Tzirita Zacharatou**, A. Kipf, I. Sabek, J. Ding, V. Markl, and A. Kemper. "Enhancing In-Memory Spatial Indexing with Learned Search". In: *Datenbanksysteme für Business, Technologie und Web (BTW)*. Gesellschaft für Informatik e.V., 2025.

[11] C. E. Muñiz-Cuza*, N. Ho, E. **Tzirita Zacharatou**, T. B. Pedersen, and B. Yang. "Spatio-temporal graph convolutional network for stochastic traffic speed imputation". In: *International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. ACM, 2022, 14:1–14:12. [Core A].

[12] M. P. Saputra*, E. **Tzirita Zacharatou**, S. Papadias, and V. Markl. "In-Place Updates in Tree-Encoded Bitmaps". In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. ACM, 2022, 18:1–18:4. [Core2021 B].

[13] E. Ouro Paz*, E. **Tzirita Zacharatou**, and V. Markl. "Towards Resilient Data Management for the Internet of Moving Things". In: *Datenbanksysteme für Business, Technologie und Web (BTW)*. Gesellschaft für Informatik e.V., 2021, pp. 279–301.

[14] E. **Tzirita Zacharatou**, A. Kipf, I. Sabek, V. Pandey, H. Doraiswamy, and V. Markl. "The Case for Distance-Bounded Spatial Approximations". In: *Conference on Innovative Data Systems Research (CIDR)*. www.cidrdb.org, 2021. [Core A].

[15] A. de Wall, B. Deiseroth, E. **Tzirita Zacharatou**, J.-A. Quiané-Ruiz, B. Demir, and V. Markl. "Agora-EO: A Unified Ecosystem for Earth Observation - A Vision for Boosting EO Data Literacy". In: *Conference on Big Data from Space*. 2021, pp. 101–104.

[16] C. Winter*, A. Kipf, C. Anneser, E. **Tzirita Zacharatou**, T. Neumann, and A. Kemper. "GeoBlocks: A Query-Cache Accelerated Data Structure for Spatial Aggregation over Polygons". In: *International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 2021, pp. 169–180. [Core A].

[17] A. Kipf, H. Lang, V. Pandey, R. A. Persa, C. Anneser, E. **Tzirita Zacharatou**, H. Doraiswamy, P. A. Boncz, T. Neumann, and A. Kemper. "Adaptive Main-Memory Indexing for High-Performance Point-Polygon Joins". In: *International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 2020, pp. 347–358. [Core A].

[18] E. **Tzirita Zacharatou**, D. Sidlauskas, F. Tauheed, T. Heinis, and A. Ailamaki. "Efficient Bundled Spatial Range Queries". In: *International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. ACM, 2019, pp. 139–148. [Core A].

[19] D. Sidlauskas, S. Chester, E. **Tzirita Zacharatou**, and A. Ailamaki. "Improving Spatial Data Processing by Clipping Minimum Bounding Boxes". In: *International Conference on Data Engineering, (ICDE)*. IEEE, 2018, pp. 425–436. [Core A*].

[20] E. **Tzirita Zacharatou**, F. Tauheed, T. Heinis, and A. Ailamaki. "RUBIK: efficient threshold queries on massive time series". In: *International Conference on Scientific and Statistical Database Management (SSDBM)*. ACM, 2015, 18:1–18:12. [Core2014 A].

Peer-Reviewed Workshop and Demonstration Publications

[21] G. Dusella, H. Gavriilidis, V. Markl, and E. **Tzirita Zacharatou**. "Benchmarking Spatial Operations over Heterogeneous Data: The Case of Zonal Statistics". In: *International Workshop on Testing Database Systems (DBTest (co-located with SIGMOD))*. ACM, 2026.

[22] H. Gavriilidis, F. Henze, J. Ziegler, J. Benn, E. **Tzirita Zacharatou**, and V. Markl. "Practical Spreadsheet Parsing with SheetReader". In: *International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 2026, pp. 756–759. Demo paper. [Core2023 A].

[23] F. Treykorn*, A. Persitzky*, M. Speer*, Z. Kaoudi, and E. **Tzirita Zacharatou**. "Cross-Platform Spatial Analytics with Apache Wayang: Geometries Have Boundaries, Your Analytics Shouldn't". In: *International Conference on Advances in Geographic Information Systems (SIGSPATIAL)*. ACM, 2026. Demo paper. [Core A].

[24] T. J. Terhaag, X. Chatziliadis*, E. **Tzirita Zacharatou**, and V. Markl. "GPU-Accelerated Stochastic Gradient Descent for Scalable Operator Placement in Geo-Distributed Streaming Systems". In: *International Workshop on Accelerating Analytics and Data Management Systems Using Modern Processor and Storage Architectures (ADMS (co-located with VLDB))*. 2025.

[25] G. Dusella*, H. Gavriilidis, L. Nuhu, V. Markl, and E. **Tzirita Zacharatou**. "Multi-Backend Zonal Statistics Execution with Raven". In: *International Conference on Management of Data (SIGMOD)*. ACM, 2024. Demo paper. [Core A*].

[26] A. Wachs* and E. **Tzirita Zacharatou**. "Analysis of Geospatial Data Loading". In: *International Workshop on Testing Database Systems (DBTest (co-located with SIGMOD))*. ACM, 2024, pp. 36–42.

[27] D. Burrell*, X. Chatziliadis*, E. **Tzirita Zacharatou**, S. Zeuch, and V. Markl. "Workload Prediction for IoT Data Management Systems". In: *Datenbanksysteme für Business, Technologie und Web (BTW)*. Gesellschaft für Informatik e.V., 2023, pp. 943–950. BTW 2023 DE4DS Workshop.

[28] A. K. Aksoy*, P. Dushev, E. **Tzirita Zacharatou**, H. Hemsen, M. Charfuelan, J. A. Quiané-Ruiz, B. Demir, and V. Markl. "Satellite Image Search in AgoraEO". In: *Proceedings of the Very Large Databases Endowment (PVLDB)* 15(12) (2022), pp. 3646–3649. Demo paper. [Core A*].

[29] F. Henze*, H. Gavriilidis, E. **Tzirita Zacharatou**, and V. Markl. "Efficient Specialized Spreadsheet Parsing for Data Science". In: *International Workshop on Design, Optimization, Languages and Analytical Processing of Big Data (DOLAP (co-located with EDBT))*. CEUR-WS.org, 2022, pp. 41–50. [Core B].

[30] X. Chatziliadis*, E. **Tzirita Zacharatou**, S. Zeuch, and V. Markl. "Monitoring of Stream Processing Engines Beyond the Cloud: An Overview". In: *Open Journal of Internet Of Things (OJIOT)* 7(1) (2021), pp. 71–82. VLDB 2021 VLIoT Workshop.

[31] S. Zeuch, E. **Tzirita Zacharatou**, S. Zhang, X. Chatziliadis, A. Chaudhary, B. D. Monte, D. Giouroukis, P. M. Grulich, A. Ziehn, and V. Markl. "NebulaStream: Complex Analytics Beyond the Cloud". In: *Open Journal of Internet Of Things (OJIOT)* 6(1) (2020), pp. 66–81. VLDB 2020 VLIoT Workshop.

[32] H. Doraiswamy, E. **Tzirita Zacharatou**, F. Miranda, M. Lage, A. Ailamaki, C. T. Silva, and J. Freire. "Interactive Visual Exploration of Spatio-Temporal Urban Data Sets using Urbane". In: *International Conference on Management of Data (SIGMOD)*. ACM, 2018, pp. 1693–1696. Demo paper. [Core A*]
Best Demonstration Award.

[33] M. Pavlovic, E. **Tzirita Zacharatou**, D. Sidlauskas, T. Heinis, and A. Ailamaki. "Space odyssey: efficient exploration of scientific data". In: *International Workshop on Exploratory Search in Databases and the Web (ExploreDB (co-located with SIGMOD))*. ACM, 2016, pp. 12–18.

Book Chapters/Entries

[34] E. **Tzirita Zacharatou**. "GPU Rasterization for Aggregate Join Queries over Polygons". In: *Encyclopedia of GIS*. Springer, 2025. To Appear.

Invited Journal Articles

[35] V. Leis, A. Gruenheid, P. Carbone, and E. **Tzirita Zacharatou**. "Reminiscences on Influential Papers". In: *SIGMOD Record* 54(3) (2025).

[36] Z. Bao, P. Bouros, R. Cheng, B. Choi, A. Dignös, W. Ding, Y. Fang, B. Han, J. Hu, A. Khan, W. Lin, X. Lin, C. Long, N. Mamoulis, J. Pei, M. Renz, S. Shekhar, J. Shi, E. **Tzirita Zacharatou**, S. Wang, X. Wang, X. Wang, R. C.-W. Wong, D. Yan, X. Yan, B. Yang, D. Yao, C. Zhang, P. Zhao, and R. Zhu. "A Summary of ICDE 2022 Research Session Panels". In: *Data Engineering Bulletin* 47 (2023).

Conference and Workshop Proceedings (Editor)

[37] E. **Tzirita Zacharatou** and A. Maté, eds. *Proceedings of the 28th International Workshop on Design, Optimization, Languages and Analytical Processing of Big Data (DOLAP 2026) co-located with the 29th International Conference on Extending Database Technology and the 28th International Conference on Database Theory (EDBT/ICDT 2026), Tampere, Finland, March 24, 2026*. Vol. 4186. CEUR Workshop Proceedings. CEUR-WS.org, 2026. Program chair.

[38] Y. Zhou, P. K. Chrysanthis, V. Gulisano, and E. **Tzirita Zacharatou**, eds. *16th ACM International Conference on Distributed and Event-based Systems, DEBS 2022, Copenhagen, Denmark, June 27 - 30, 2022*. ACM, 2022. ISBN: 978-1-4503-9308-9. Proceedings chair.

PhD Thesis

[39] E. **Tzirita Zacharatou**. "Efficient Query Processing for Spatial and Temporal Data Exploration". PhD thesis. EPFL, 2019, p. 183.